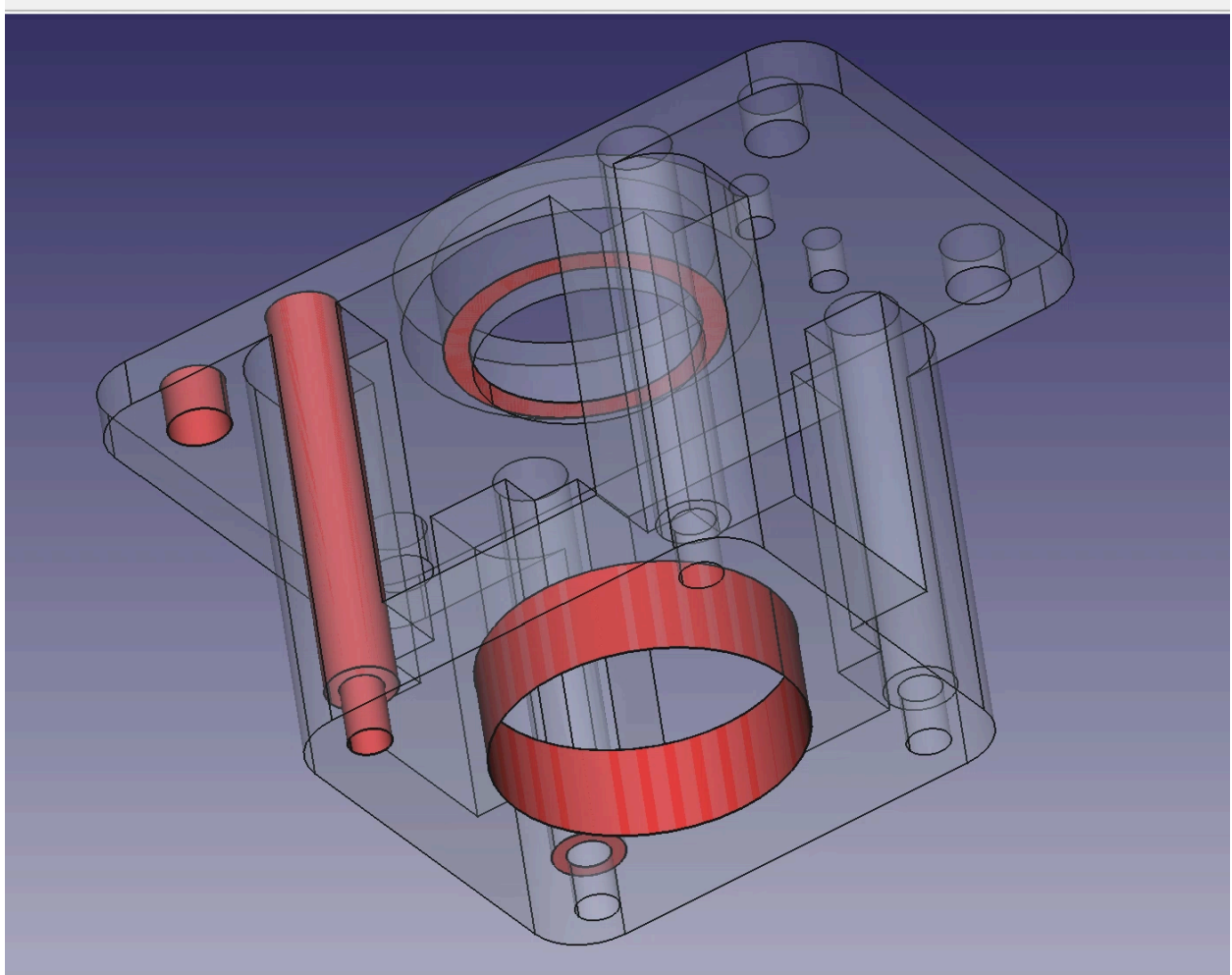


ML Engineer - Take-home

Feature Recognition Across CAD Models

Task

Given a reference STEP model + a labeled feature in it (face IDs + OCC face IDs), find all occurrences of that feature in a set of query STEP models. "Same feature" is semantic - a counterbored hole should match counterbored holes regardless of size, orientation, or surrounding geometry.



Output per query model: list of instances, each a set of face IDs + OCC face IDs, with a confidence score.

Approach

Your call. Contrastive on B-Rep graphs, graph matching, retrofitting a B-Rep encoder (UV-Net, BRepNet, AAGNet, SB-GCN, etc.), an LLM over a structured serialization, vision models on rasterized views, point clouds — pick what fits and justify it.

If you use an LLM, we want to see you've thought about the serialization, token cost per query, and what happens when a model exceeds context.

Supervised is fine for V1. Tell us what the self-supervised path looks like.

Extensibility (non-blend paradigm)

At Hanomi we focus on non-blend features — holes, pockets, slots, keyways, bosses — and the framework is built so adding a new feature type is mostly a data problem, not an architecture one.

Train on one feature type. Then add a second and report what changed — data, training, inference, accuracy. Tell us what breaks at the 10th or 50th.

Evaluation

Roughly weighted:

1. Problem framing
2. Representation choice for B-Rep
3. Context / token / compute budgeting
4. Benchmarking: at least one baseline, ablations on the choices that matter
5. The extensibility demo above
6. Engineering taste — clean, reproducible

Not evaluating: SOTA numbers, UI polish, long write-ups. A thoughtful 60% beats a black-box 90%.

Bonus

- Retrofitting an approach from a different domain (protein contact, molecular similarity, point cloud registration, program synthesis, whatever) — argue honestly about the trade-offs
- Working self-supervised angle, even on a subset
- Cost analysis: \$/query or GPU-seconds/query at production scale
- Failure-mode analysis - where does it break

Datasets

MFCAD / MFCAD++ is closest. Also Fusion 360 Gallery, ABC. Synthetic is fine if you explain generation. Use `pythonocc-core` for STEP parsing.

Deliverables

1. Private GitHub repo, README with run instructions
2. Lots of results on how your model performs for diff features across different step files, programmatic way of you documenting them

Timeline

5 days part-time.

Submitting

Repo + write-up. 45-min walkthrough after we read it — expect us to push on your choices.

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